

Experiment 10: Expectation Maximization (EM) Algorithm

Aim

To implement the Expectation Maximization (EM) algorithm using Gaussian Mixture Models (GMM) for clustering the Iris dataset.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the Iris dataset.

Step 3: Create a Gaussian Mixture Model (GMM) with the required number of clusters.

Step 4: Train the GMM using the dataset.

Step 5: Predict the cluster labels for all data samples.

Step 6: Display the assigned cluster labels.

Program

```
#Experiment 10: Expectation Maximization (EM) Algorithm
from sklearn.datasets import load_iris
from sklearn.mixture import GaussianMixture

iris = load_iris()

X = iris.data

gmm = GaussianMixture(
    n_components=3,
    random_state=0
)

gmm.fit(X)

labels = gmm.predict(X)

print("Cluster Labels:")
print(labels)
```

Output

```
*** Cluster Labels:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 2 0 2 0
 0 0 0 2 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2]
```

Result

Thus, the Expectation Maximization (EM) algorithm was successfully implemented using the Gaussian Mixture Model, and the Iris dataset was clustered into three groups successfully.